

Polygenic prediction and sparse regression analysis

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Today's lecture

- 1 What is GWAS (review and pick up the leftover)?
- 2 What is a polygenic score model?
- 3 What are the statistical challenges in PGS estimation?
- 4 Statistical fine-mapping to handle LD structures

What is GWAS (review and pick up the leftover)?

Using the 1000 Genomes data via bigsnpr

```
library(bigsnp)  
download_1000G("../data/genotype/")
```

```
data <- snp_attach(.bk.file)  
str(data, max.level=1, strict.width = "cut")
```

```
## List of 3  
## $ genotypes:Reference class 'FBM.code256' [package "bigsnpr"]  
## ..and 26 methods, of which 12 are possibly relevant  
## $ fam      : 'data.frame':   2490 obs. of  6 variables:  
## $ map      : 'data.frame':  1664852 obs. of  6 variables:  
## - attr(*, "class")= chr "bigSNP"
```


Notations

- $X: n \times p$ genotype matrix of n samples/individuals and p variants
- $Y: n \times 1$ phenotype vector of n samples/individuals

Let's do GWAS to find the disease-associated variants

A variant-by-variant association t-test for a variant j :

$$H_0 : \quad \mathbb{E}[X_{ij}|Y_i = 1] = \mathbb{E}[X_{ij}|Y_i = 0]$$

vs.

$$H_1 : \quad \mathbb{E}[X_{ij}|Y_i = 1] \neq \mathbb{E}[X_{ij}|Y_i = 0]$$

- The average genotype is the same between the case and control under the null.

For one variant's test

```
j <- 1
```

```
t.test(X[Y == 0, j], X[Y == 1, j])
```

```
##
```

```
##  Welch Two Sample t-test
```

```
##
```

```
## data:  X[Y == 0, j] and X[Y == 1, j]
```

```
## t = 0.16273, df = 2468.6, p-value = 0.8707
```

```
## alternative hypothesis: true difference in means is not e
```

```
## 95 percent confidence interval:
```

```
##  -0.02015587  0.02380397
```

```
## sample estimates:
```

Can we calculate GWAS statistics more efficiently?

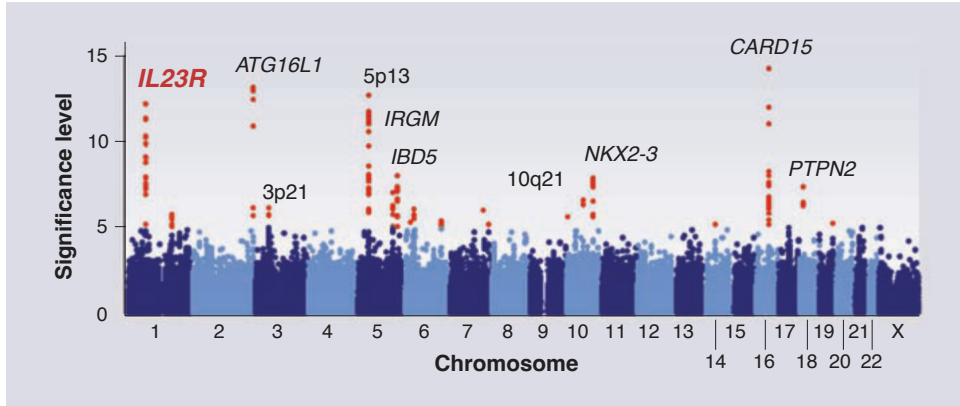
```
## library(matrixTests)
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ])
```

- For each pair of columns in the two matrices, the function performs t-test and summarize all the results into a list of vectors.

```
names(.gwas)
```

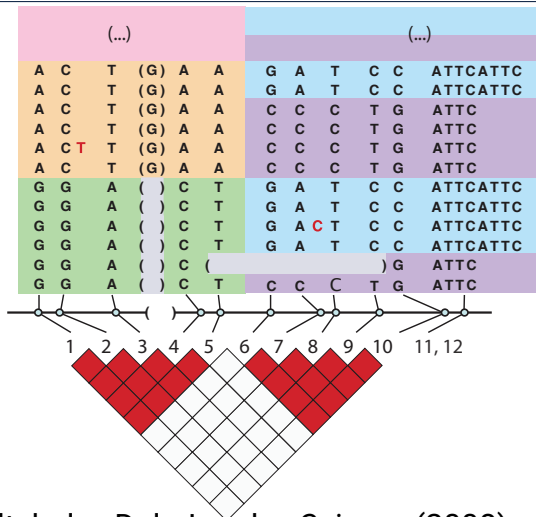
```
## [1] "obs.x"      "obs.y"      "obs.tot"    "mean.x"     "mean.y"
## [6] "mean.diff"  "var.x"      "var.y"      "stderr"     "df"
## [11] "statistic"  "pvalue"     "conf.low"   "conf.high"  "mean.null"
## [16] "alternative" "conf.level"
```

Human genetic variation recap



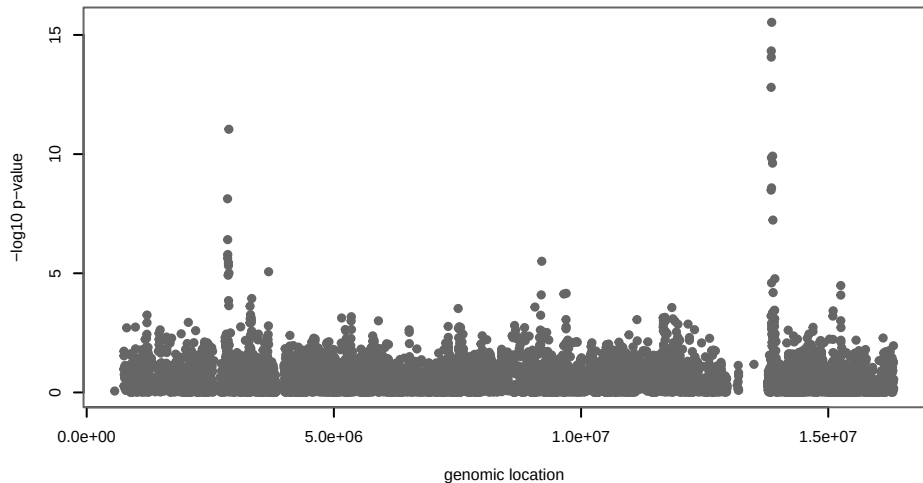
Highly recommend Altshuler, Daly, Lander, *Science* (2008)

Human genetic variation recap

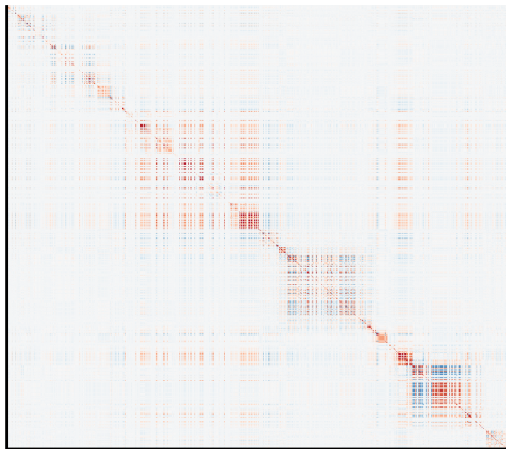


Highly recommend Altshuler, Daly, Lander, *Science* (2008)

Manhattan plot: a summary of all the GWAS p-values



The covariance of X between the variants (columns)



For a genotype matrix X ($n \times p$),

Linkage disequilibrium matrix

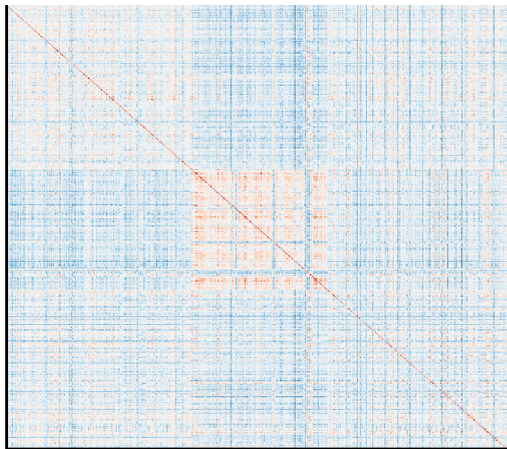
A $p \times p$ matrix:

$$\hat{R} = \frac{1}{n} X^\top X$$

given each column x_j standardized
(mean 0, SD 1).

$$R_{ij} = \frac{1}{n} \sum_{r=1}^n X_{ri} X_{rj}$$

The covariance of X between the individuals (rows)



For a genotype matrix X ($n \times p$)

Genetic relatedness matrix

An $n \times n$ matrix:

$$\hat{K} = \frac{1}{p} X X^\top$$

given each row x_i standardized (mean 0, SD 1).

$$K_{ij} = \frac{1}{p} \sum_{g=1}^p X_{ig} X_{jg}$$

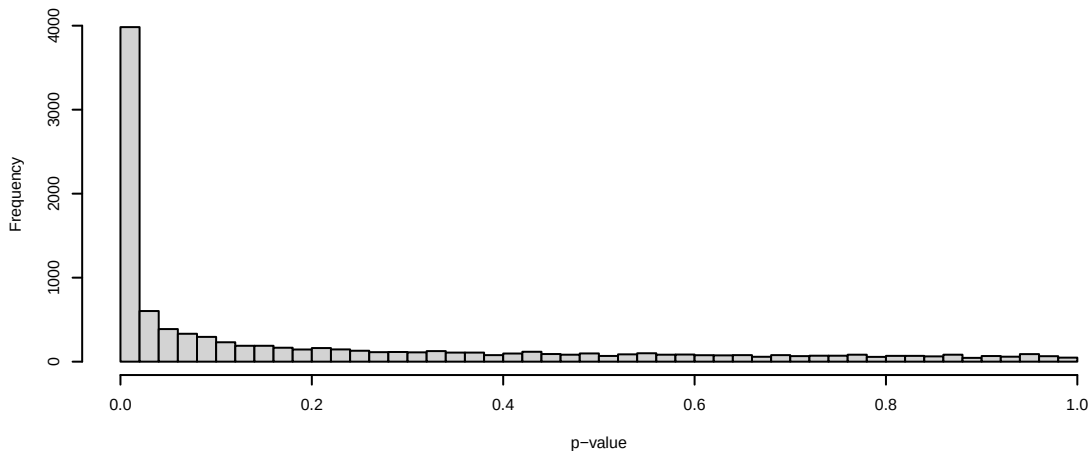
What would happen if there were AG-specific biases?

$$g \left(\begin{matrix} Y \\ \text{phenotype} \end{matrix} \right) \sim \underbrace{\sum_{k \in \text{causal variants}} X_k \beta_k}_{\text{true genetic effect}} + \underbrace{\sum_g U_g \gamma_g}_{\text{ancestry group factors}} + \epsilon$$

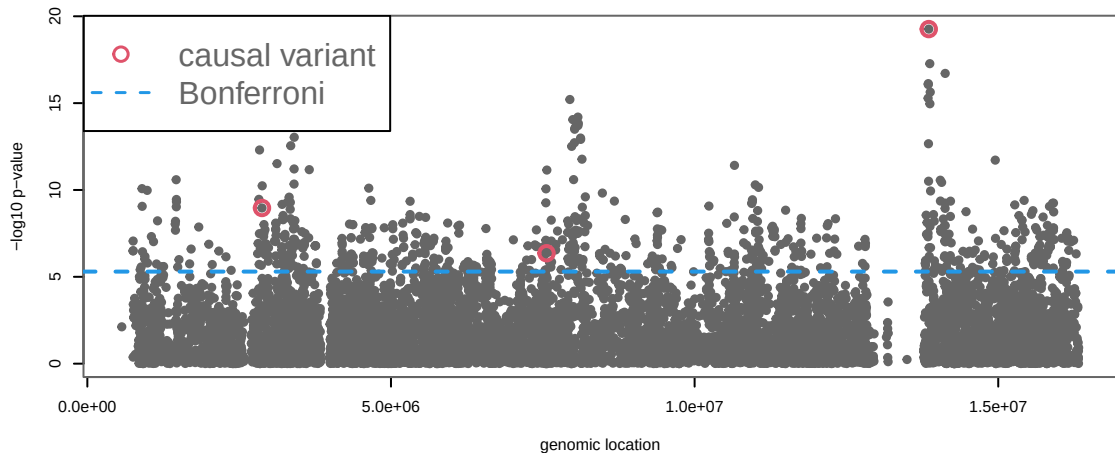
Do GWAS as usual by variant-by-variant t-test:

```
.gwas.biased <- col_t_welch(X[Y == 0, ], X[Y == 1, ])
```

Okay... do we have that many GWA-significant variants?



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How do we know there is an unknown group structure?

GWAS assumption:

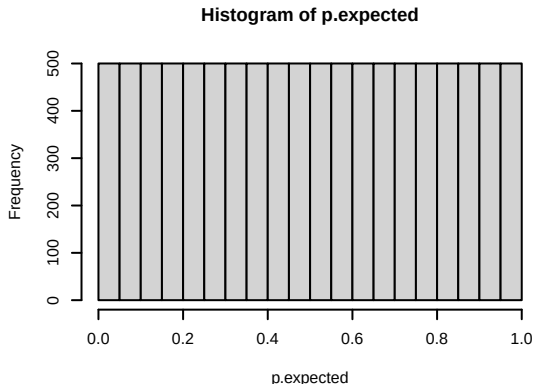
- Only a small portion of variants are significantly associated. Why?
- A majority of p-values under the null will follow Uniform distribution.
- What do we expect from Uniform p-value?

```
nn <- length(.gwas$pvalue)
p.expected <- seq(1, nn)/(nn+1)
```

How do we know there is an unknown group structure?

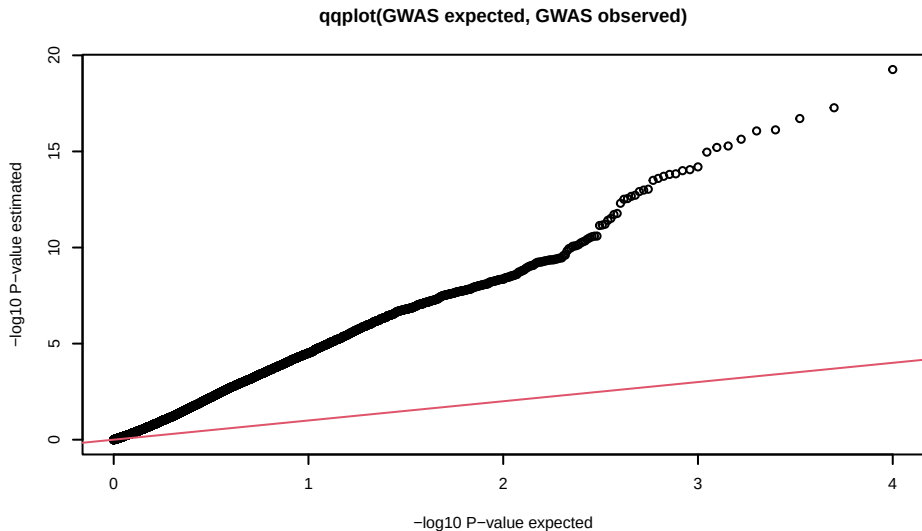
GWAS assumption:

- Only a small portion of variants are significantly associated. Why?
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```
nn <- length(.gwas$pvalue)
p.expected <- seq(1, nn)/(nn+1)
```

However, when we draw quantile-quantile



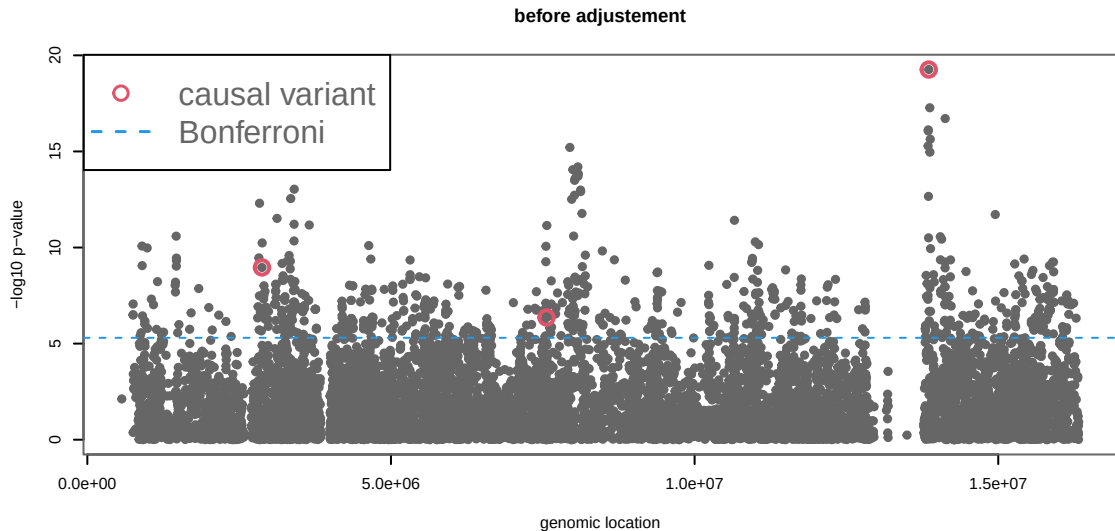
Can we quickly fix this? A little bit...

```
.svd <- rsvd::rsvd(X, 10)
Y.adj <- lm(Y ~ .svd$u)$residual
.gwas.adj <- col_cor_pearson(X, Y.adj)
```

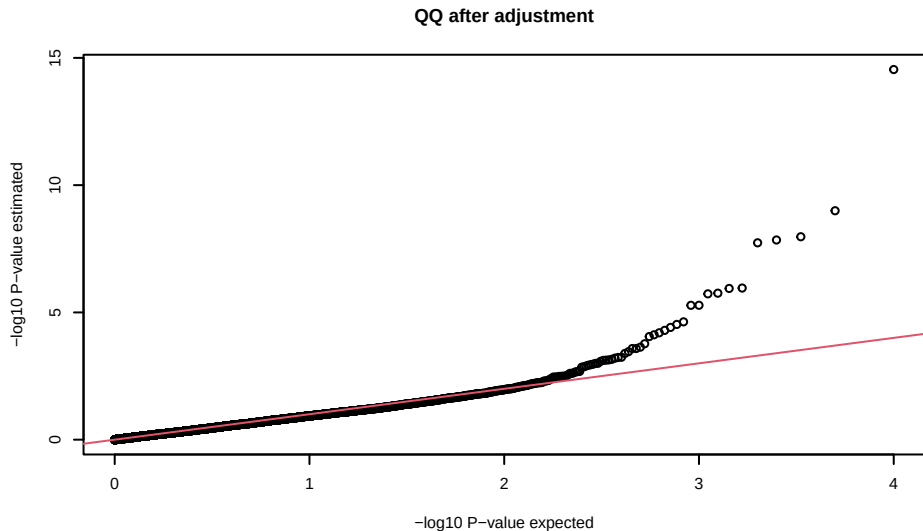
Assuming

$$g \begin{pmatrix} Y \\ \text{phenotype} \end{pmatrix} \sim \underbrace{X_k \beta_k}_{\text{true genetic effect}} + \underbrace{\sum_g u_g \gamma_g}_{\text{estimate this by SVD}} + \epsilon$$

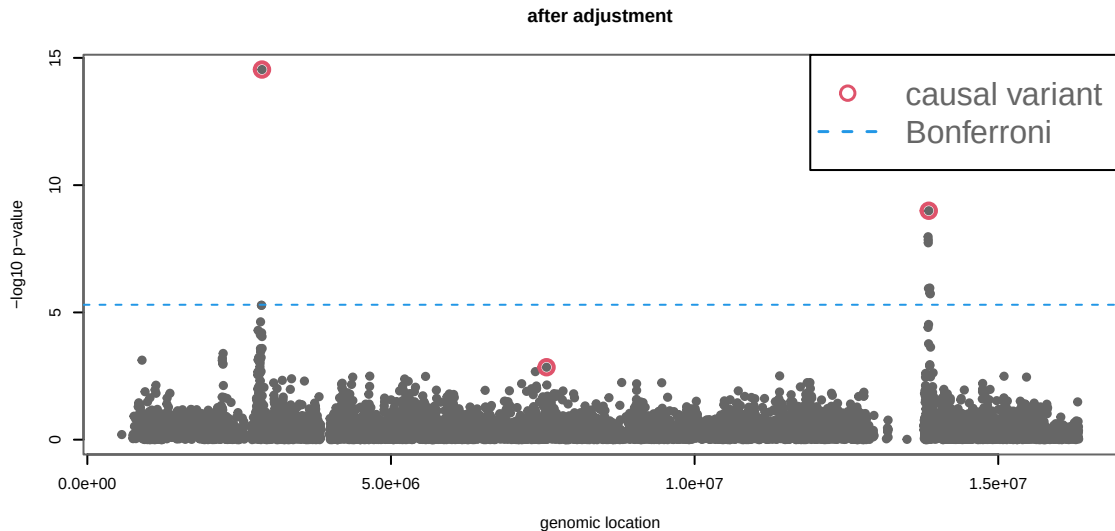
Simply regressing out population PCs can reduce biases



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Linear Mixed Effect model: a better way to adjust pop struct

A linear regression model:

$$y = \mathbf{x}_j \beta_j + \epsilon$$

a fixed genetic effect

What are we missing? Can we assume homo-scedasticity, i.e.,

$$\epsilon \stackrel{?}{\sim} \mathcal{N}(0, \sigma^2 I)$$

Linear Mixed Effect model: a better way to adjust pop struct

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What are we missing? Can we assume homo-scedasticity, i.e.,

$$\epsilon \stackrel{?}{\sim} \mathcal{N}(0, \sigma^2 I)$$

A linear model with a random effect:

$$y = \mathbf{x}_j \beta_j + \mathbf{u} + \epsilon$$

fixed random effect

Note: Now, we assume:

$$\epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

A linear model with population-driven random effects - 2

- Unwanted random effects to **remove**: $n \times 1$ vector $\mathbf{u}$
- **Goal**: the fixed genetic effect of a j .

$$y = \mathbf{x}_j \underset{\text{goal}}{\beta_j} + \underset{\text{remove}}{\mathbf{u}} + \epsilon$$

A linear model with population-driven random effects - 2

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- 1 Note that $\mathbf{u}$ shouldn't be tied to a particular variant (by definition)

A linear model with population-driven random effects - 2

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$$y = \underset{\text{goal}}{\mathbf{x}_j \beta_j} + \underset{\text{remove}}{\mathbf{u}} + \epsilon$$

- 1 Note that $\mathbf{u}$ shouldn't be tied to a particular variant (by definition)
- 2 Also, the covariation of $\mathbf{u}$ is primarily driven by relatedness among individuals, not the variants.

$$\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \tau^2 K), \quad K \approx \frac{1}{n} X X^\top$$

LMM to estimate unbiased GWAS, removing random u

A hierarchical model:

$$y|X, \beta, u, \sigma \sim \mathcal{N}(X\beta + u, \sigma^2 I) \quad (1)$$

$$u|\tau, K \sim \mathcal{N}(0, \tau^2 K) \quad (2)$$

If we integrate out u ,

$$y|X, \beta \sim \mathcal{N}\left(y \middle| X\beta, \underbrace{\tau^2 K}_{\text{genetic-relatedness matrix}} + \underbrace{\sigma^2 I}_{\text{irreducible}}\right)$$

How did we handle high-dimensionality?

A key question in LMM: What covariance matrix?

If there were many types of random effects,

$$y = \underset{\text{fixed}}{X\beta} + \underset{\text{random effects}}{u + w + \dots} + \underset{\text{unknown}}{\epsilon}$$

A key question in LMM: What covariance matrix?

$$y = \underbrace{X\beta}_{\text{fixed}} + \underbrace{u + w + \dots}_{\text{random effects}} + \underbrace{\epsilon}_{\text{unknown}}$$

We would need to many covariance matrices:

$$y|\cdot \sim \mathcal{N}\left(X\beta, \sigma^2(I + \underbrace{\delta_u K_u + \delta_w K_w + \dots}_{\text{random effects}})\right)$$

A key question in LMM: What covariance matrix?

$$y = \underbrace{X\beta}_{\text{fixed}} + \underbrace{u + w + \dots}_{\text{random effects}} + \underbrace{\epsilon}_{\text{unknown}}$$

We would need to many covariance matrices:

$$y|\cdot \sim \mathcal{N}\left(X\beta, \sigma^2(I + \underbrace{\delta_u K_u + \delta_w K_w + \dots}_{\text{random effects}})\right)$$

If we only care about variance decomposition $\beta_j \sim \mathcal{N}(0, \tau)$:

$$y \sim \mathcal{N}\left(0, \sigma^2\left(\underbrace{\frac{\sigma_{\text{genetic}}^2}{n}XX^\top}_{\text{observed genetic}} + I + \underbrace{\delta_u K_u + \delta_w K_w + \dots}_{\text{random effects}}\right)\right)$$

Should we worry about “over-fitting” in LMM?

An equivalent question for PCA-based confounder adjustment:

How many PCs to adjust in GWAS?

Can we include a candidate SNP in the GRM K matrix?

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An equivalent question for PCA-based confounder adjustment:

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Can we include a candidate SNP in the GRM K matrix?

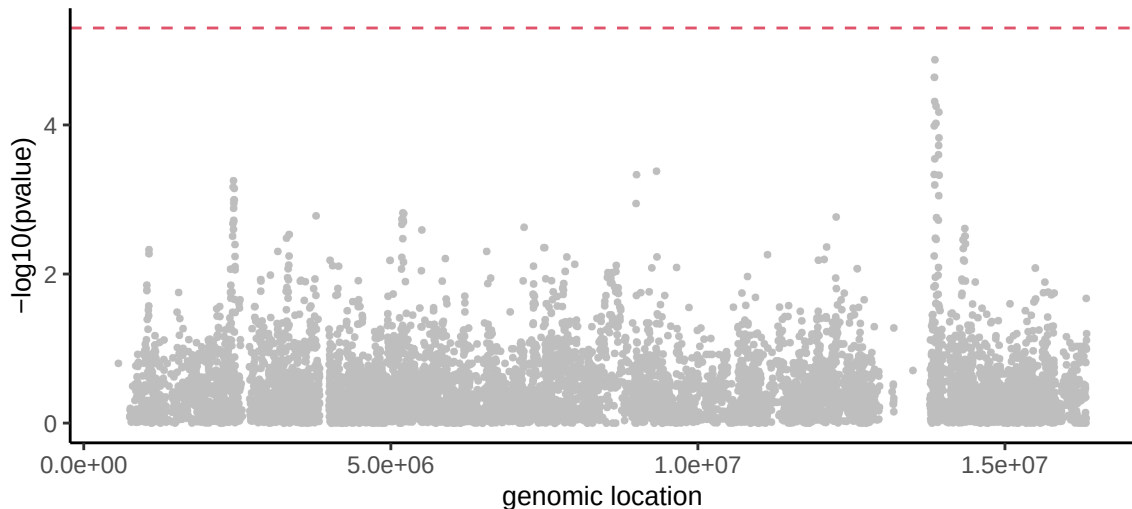
For each chromosome $c \in \{1, \dots, 22, X, Y\}$, build a leave-one-chromosome-out (LOCO) kinship matrix, say K_{-c} :

$$\mathcal{N}(y | x_j \beta_j, \sigma^2 (\delta K_{-c} + I))$$

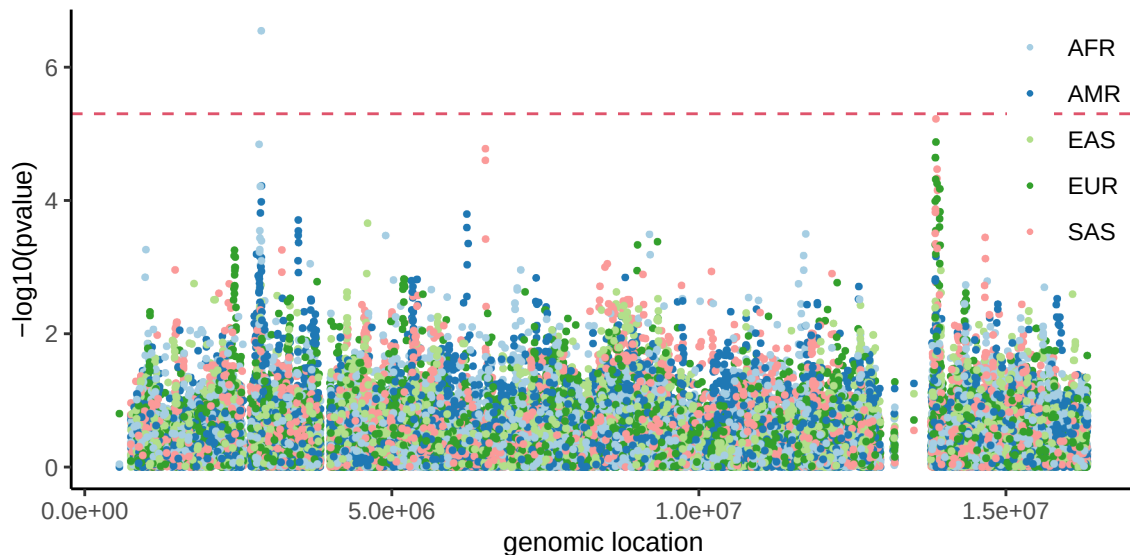
Yang, et al., *Nature Genetics* (2014); Tucker, Price, Berger, *Genetics* (2014)

GWAS summary statistics within each pop

GWAS in EUR (N=503)



GWAS summary statistics within each pop



Meta analysis: How can we combine multiple studies?

For each population/study k :

- $\hat{\beta}_{jk}$: effect size for a variant j on a study k
- $s_{jk} = \text{SE}(\beta_{jk})$: standard error for a variant j on a study k
- $w_{jk} = 1/s_{jk}^2$: inverse variance

Meta analysis by inverse variance weighting

$$\hat{\beta}_j = \frac{\sum_k w_k \hat{\beta}_{jk}}{\sum_k w_k}, \quad \text{SE}(\hat{\beta}_j) = \frac{1}{\sqrt{\sum_k w_k}}$$

Meta analysis can boost statistical power

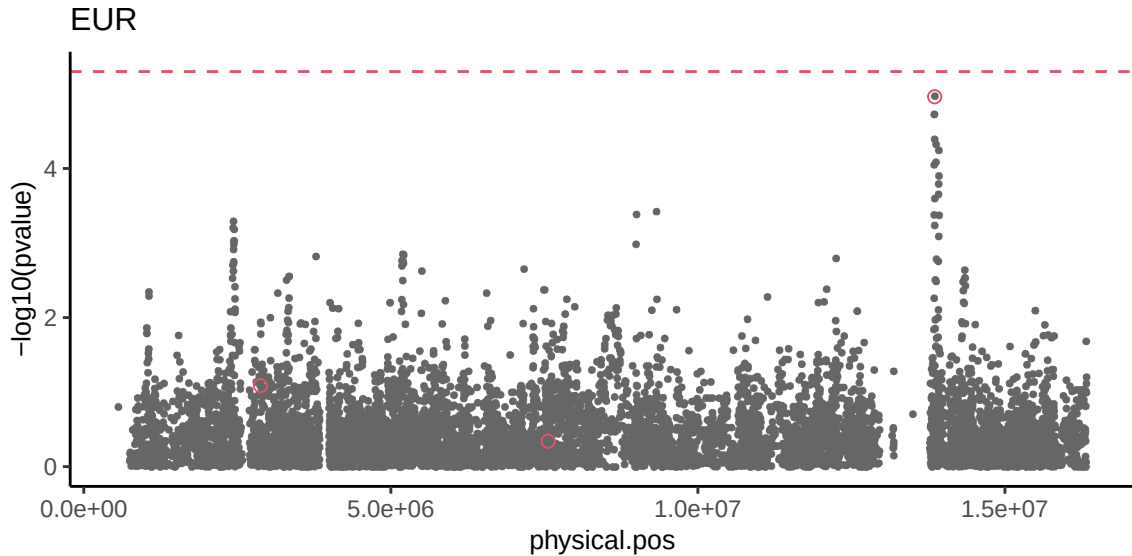
```
gwas.dt[, w := 1/(`stderr`^2 + 1e-16)]
```

```
meta.dt <-
```

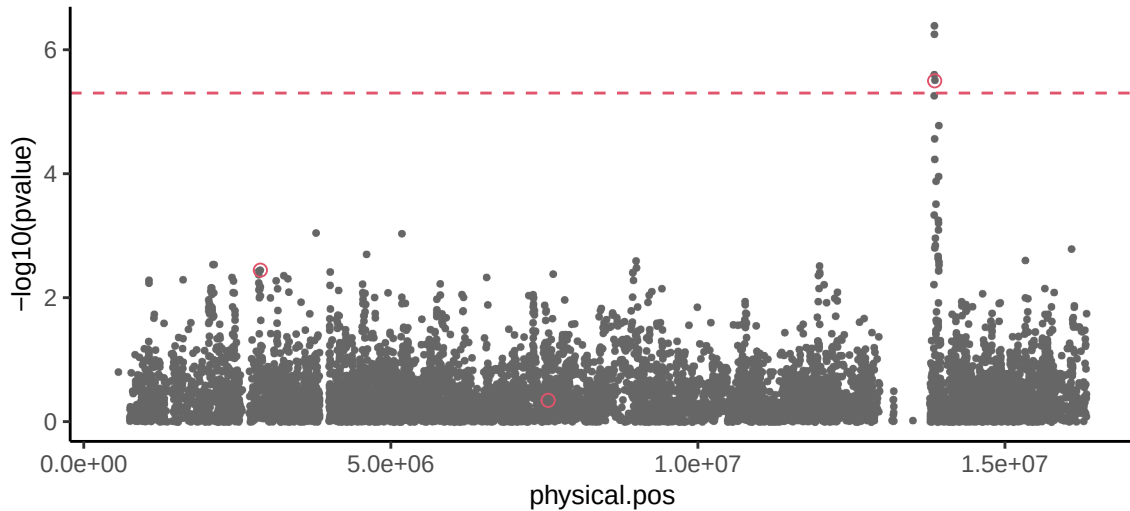
```
  gwas.dt[!is.na(w),  
    .(`mean` = sum(`mean.diff` * w)/sum(w),  
      `se` = 1/sqrt(sum(w))),  
    by = .(physical.pos)]
```

```
meta.dt[, `z` := `mean`/`se`]
```

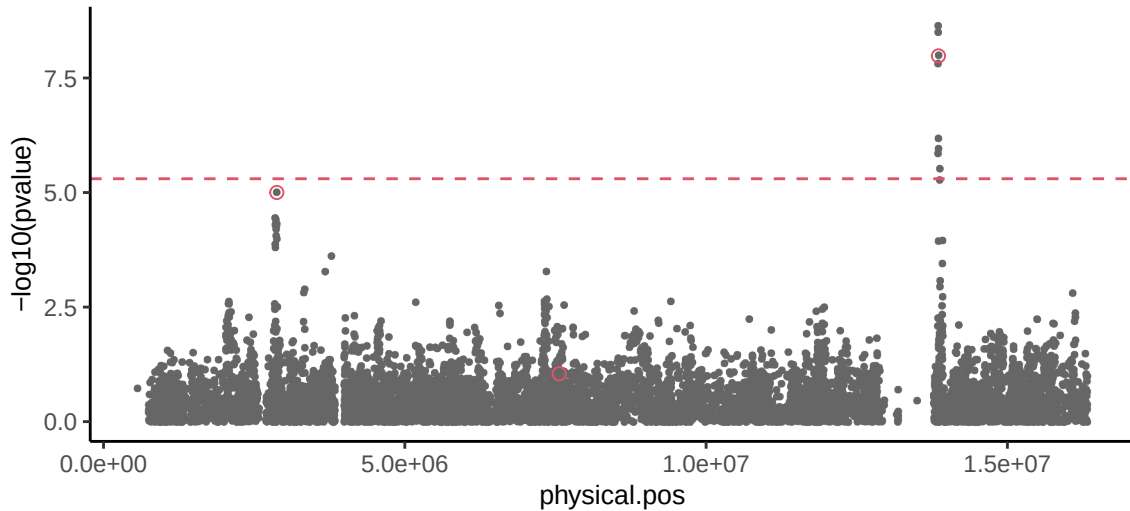
```
meta.dt[, `pvalue` := 2*pnorm(abs(`z`), lower.tail=F)]
```



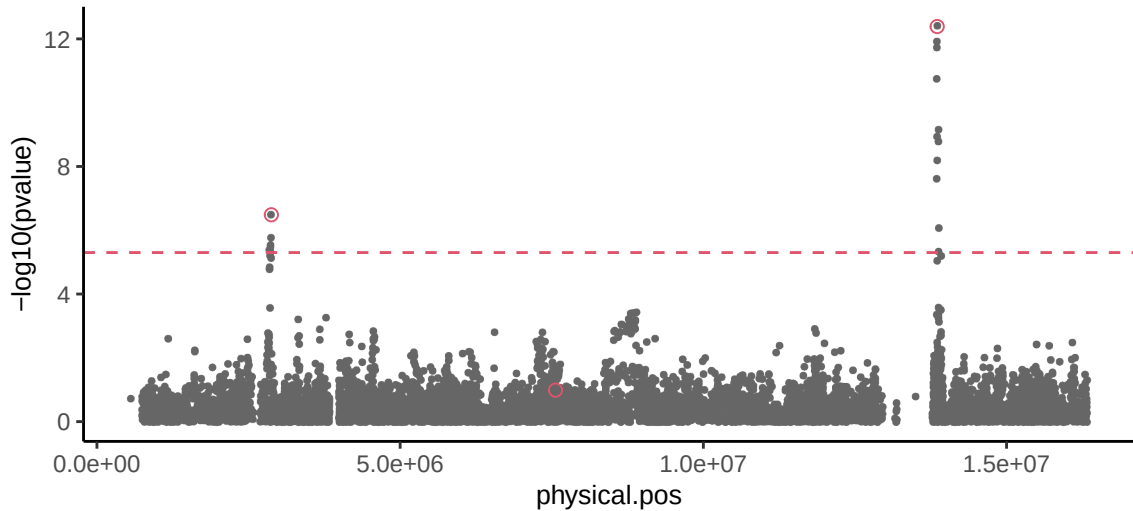
EUR + EAS



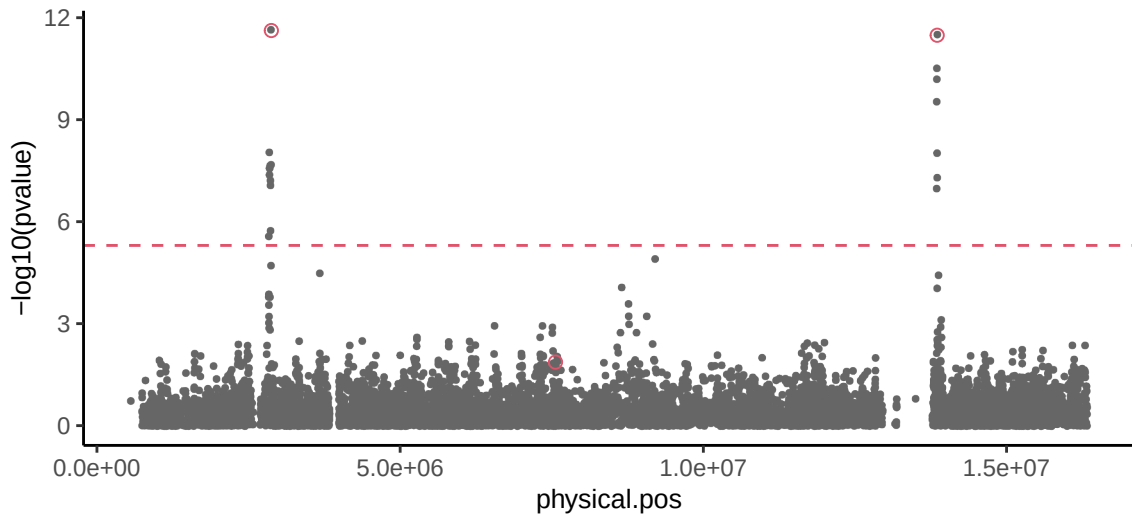
EUR + EAS + AMR



EUR + EAS + AMR + SAS



EUR + EAS + AMR + SAS + AFR



What's next after GWAS?

- GWAS until 2010s: heavy focuses on mapping
 - GWAS map: genetic variants $\rightarrow$ a phenotype
 - Stringent genome-wide p-value cutoff
 - Study design, meta analysis
- NHGRI-EBI GWAS Catalog:
<https://www.ebi.ac.uk/gwas/>

Let's take a look at the GWAS Catalog:
<https://www.ebi.ac.uk/gwas/>

What's next after GWAS?

- GWAS since 2010s: more emphases on prediction
 - Can we turn GWAS results to a prediction model?
 - Can we understand the mechanisms?
 - Machine learning, data integration, causal inference

Today's lecture

- 1 What is GWAS (review and pick up the leftover)?
- 2 What is a polygenic score model?**
- 3 What are the statistical challenges in PGS estimation?
- 4 Statistical fine-mapping to handle LD structures

What is a polygenic score model?

Polygenic score models predict disease prevalence based on genome

- Poly + genic = many genes (units of hereditary components)

A (linear) polygenic score (PGS)

$$Y_i = \sum_j X_{ij} \beta_j$$

for an individual i .

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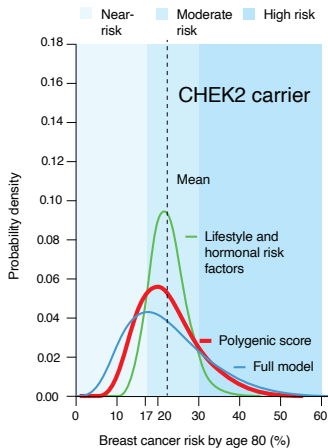
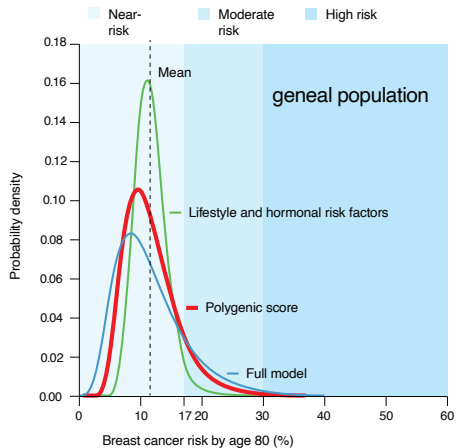
$$Y_i = \sum_j X_{ij} \beta_j$$

for an individual i .

Knowing β 's, we can predict:

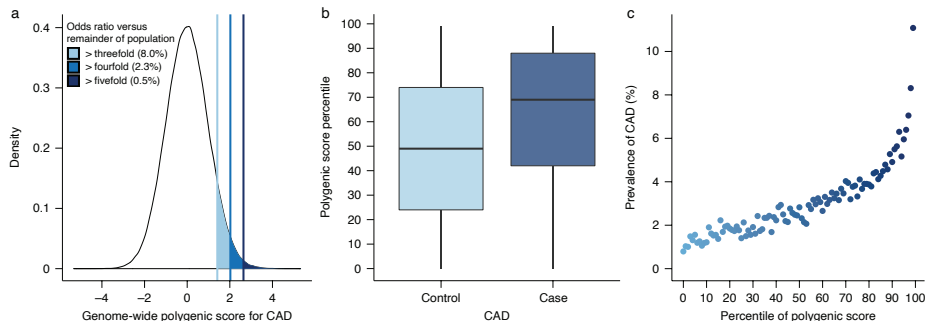
$$Y^*_i \leftarrow \sum_j X^*_{ij} \hat{\beta}_j$$

Example: PGS for breast cancer occurrence



- Strong heritability (proportion of disease risk variation explained by genetics): 35% - 80%
- Prediction more accurate than life style and other environmental factors
- More powerful if combined with rare risk factors

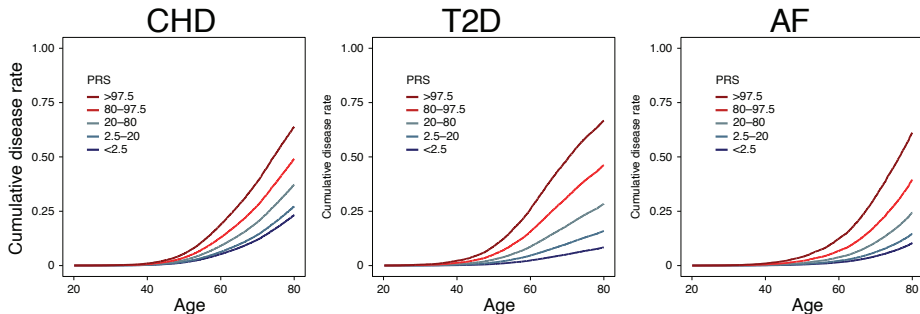
Example: PGS for coronary artery disorder → prevention



● Top .5% of PGS values → five fold increase of CAD

Khera .. Kathiresan, Nature Genetics (2018)

Example: PGS stratifies individuals' disease onset & risk

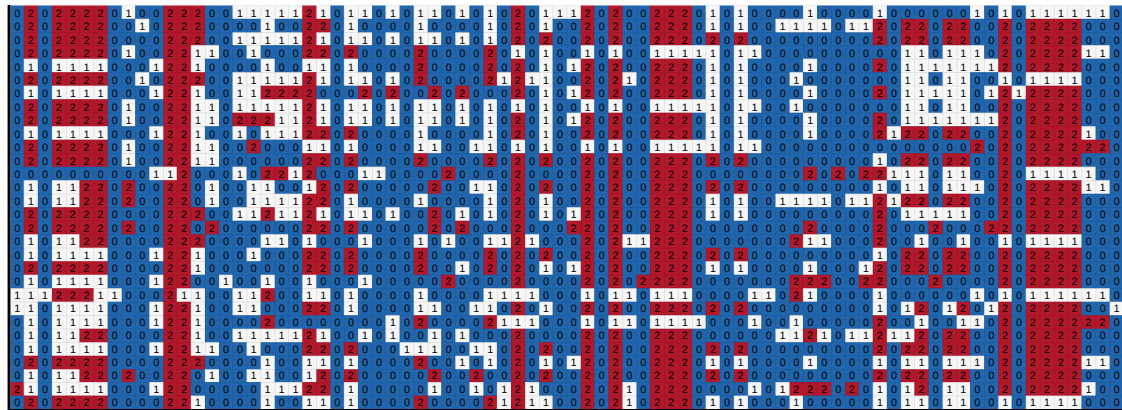


- PRS $\approx$ PGS.
- We can partition cohorts based on PGS profiles.

Mars ... Ripatti, Nature Medicine (2020)

A toy example: GWAS for “Obsessive ggplot Disorder”

A genotype matrix X ($X_{ij} \in \{0, 1, 2\}$):



¹just an illustration purpose

In GWAS, we can have case vs. control phenotype

$Y_i = 1$ if case vs. $Y_i = 0$ if control

For our *OGD* GWAS:

```
table(y)
```

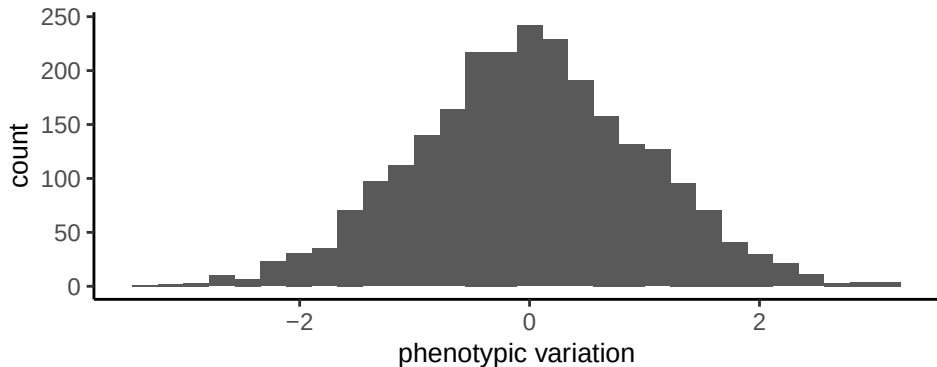
```
## y
```

```
##    0    1
```

```
## 731 769
```

- E.g., Cancer vs. non-cancer, schizophrenia vs. no mental disorder
- The fundamental question is, “Which individuals/subjects can be truly labelled control/wild type?”

The under the hood, there are quantitative “risk” scores



- E.g., height, BMI, parents' age of death, how man pack of cigarettes, etc.

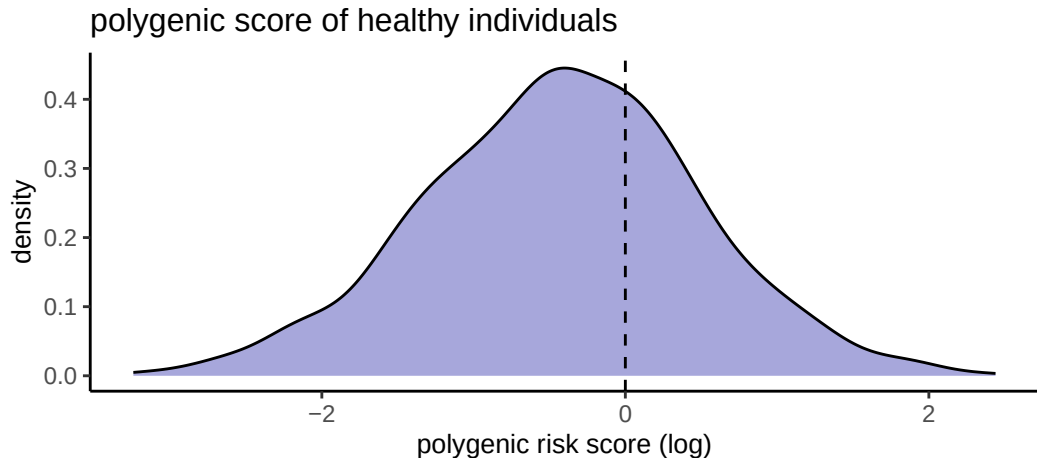
Side note: Almost all complex traits are polygenic

Polygenic effects of common SNPs → polymorphism in phenotypes

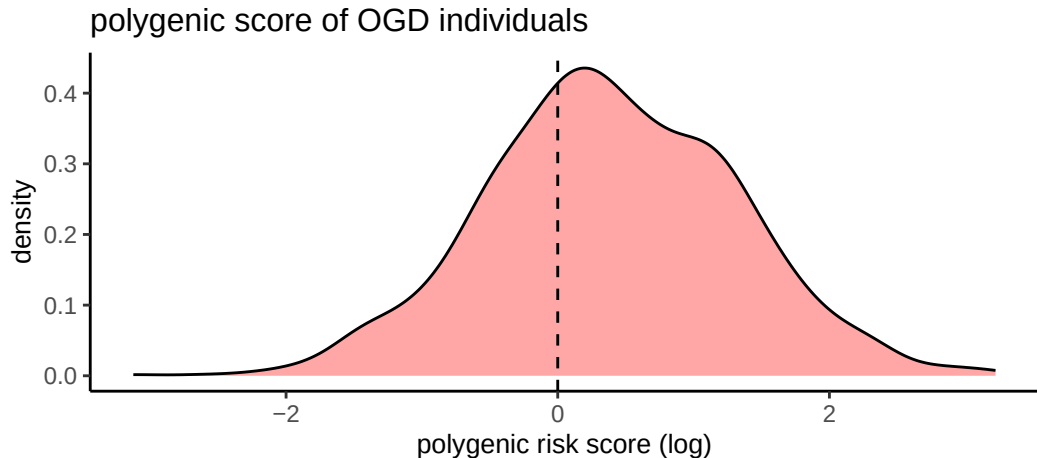
A polygenic trait is a characteristic, such as height or skin color, that is influenced by two or more genes. Because multiple genes are involved, polygenic traits do not follow the patterns of Mendelian inheritance. Many polygenic traits are also influenced by the environment and are called multifactorial.

<https://www.genome.gov/genetics-glossary/Polygenic-Trait>

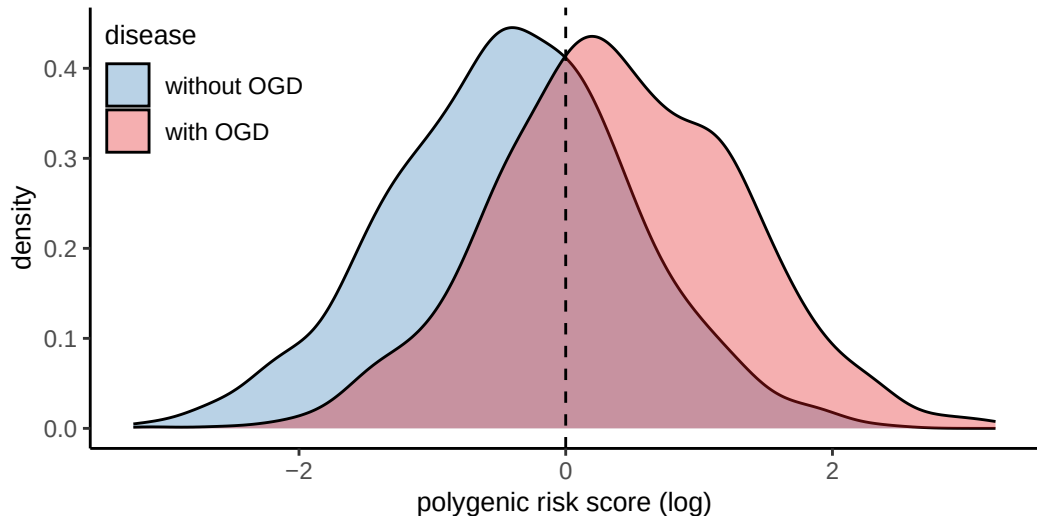
A polygenic score to predict disease prevalence



A polygenic score to predict disease prevalence



A polygenic score to predict disease prevalence



Polygenic score $\propto$ the odds of the case vs. control in GWAS

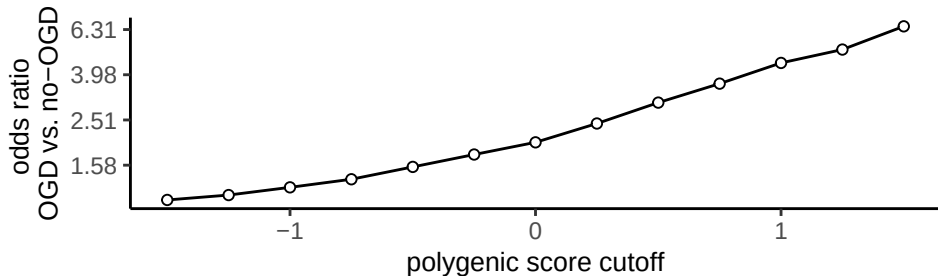
Log-odds ratio:

$$g(y) = \log \frac{p(\text{disease} | Y > y)}{p(\text{no disease} | Y > y)}$$

Polygenic score $\propto$ the odds of the case vs. control in GWAS

Log-odds ratio:

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A PGS model to explain disease prevalence with genetics

Log-odds ratio:

$$g(y) = \log \frac{p(\text{disease} | Y > y)}{p(\text{no disease} | Y > y)}$$

Goal: Estimate a function to predict this log-odds ratio.

$$g(y_i) \sim \beta_1 X_{i1} + \beta_2 X_{i2} \cdots + \epsilon$$

A PGS model to explain disease prevalence with genetics

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Goal: Estimate a function to predict this log-odds ratio.

$$g(y_i) \sim \underbrace{\beta_1 X_{i1} + \beta_2 X_{i2} \cdots}_{\text{genetic effects}} + \epsilon$$

PGS estimation is a linear modelling with $p \gg n$

$$Y_i \sim X_{i1}\beta_1 + X_{i2}\beta_2 + \dots$$

PGS estimation is a linear modelling with $p \gg n$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \sim \begin{pmatrix} X_{11} \\ X_{21} \\ \vdots \\ X_{n1} \end{pmatrix} \beta_1 + \begin{pmatrix} X_{12} \\ X_{22} \\ \vdots \\ X_{n2} \end{pmatrix} \beta_2 + \dots$$

PGS estimation is a linear modelling with $p \gg n$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \sim \begin{pmatrix} X_{11} \\ X_{21} \\ \vdots \\ X_{n1} \end{pmatrix} \beta_1 + \begin{pmatrix} X_{12} \\ X_{22} \\ \vdots \\ X_{n2} \end{pmatrix} \beta_2 + \cdots + \begin{pmatrix} X_{1p} \\ X_{2p} \\ \vdots \\ X_{np} \end{pmatrix} \beta_p$$

PGS estimation is a linear modelling with $p \gg n$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \sim \begin{pmatrix} X_{11} \\ X_{21} \\ \vdots \\ X_{n1} \end{pmatrix} \beta_1 + \begin{pmatrix} X_{12} \\ X_{22} \\ \vdots \\ X_{n2} \end{pmatrix} \beta_2 + \cdots + \begin{pmatrix} X_{1p} \\ X_{2p} \\ \vdots \\ X_{np} \end{pmatrix} \beta_p$$

$$\mathbf{y} \sim \mathbf{x}_1 \beta_1 + \mathbf{x}_2 \beta_2 + \cdots + \mathbf{x}_p \beta_p$$

$$p \gg n$$

PGS estimation is a linear modelling with $p \gg n$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \sim \begin{pmatrix} X_{11} \\ X_{21} \\ \vdots \\ X_{n1} \end{pmatrix} \beta_1 + \begin{pmatrix} X_{12} \\ X_{22} \\ \vdots \\ X_{n2} \end{pmatrix} \beta_2 + \cdots + \begin{pmatrix} X_{1p} \\ X_{2p} \\ \vdots \\ X_{np} \end{pmatrix} \beta_p$$

$$\mathbf{y} \sim \mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2 + \cdots + \mathbf{x}_p\beta_p$$

$$p \gg n$$

$n \approx 10^4$ but $p \approx 10^6$ for many large-scale GWAS

$p \gg n$: Why can't we just fit a model?

```
## lm.out <- lm(Y ~ X)
```

- $p \gg n$: need to estimate $\beta_1, \dots, \beta_p$

$p \gg n$: Why can't we just fit a model?

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## lm.out <- lm(Y ~ X)
```

- $p \gg n$: need to estimate $\beta_1, \dots, \beta_p$
- How many samples? How many unknowns?

$p \gg n$: Why can't we just fit a model?

```
## lm.out <- lm(Y ~ X)
```

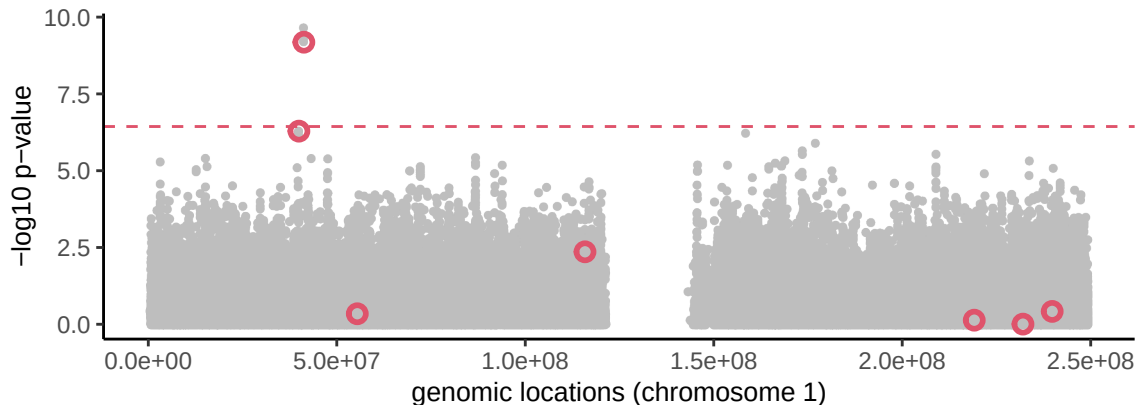
- $p \gg n$: need to estimate $\beta_1, \dots, \beta_p$
- How many samples? How many unknowns?
- Is it computationally feasible?

$p \gg n$: Variant-by-variant GWAS is not a bad idea

```
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ]) # univar.
```

$p \gg n$: Variant-by-variant GWAS is not a bad idea

```
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ]) # univar.
```



A linear model including only these “GWAS” variants

```
X.gwas <- X[, p.adjust(.gwas$lm.out <- lm(Y ~ X.gwas)
head(X.gwas, 3)
```

```
##           41270163 41270937 41272533 41274422
## HG00159           0         0         0         0
## HG01495           1         1         1         1
## HG01519           2         2         2         2
```

```
anova(lm.out)
```

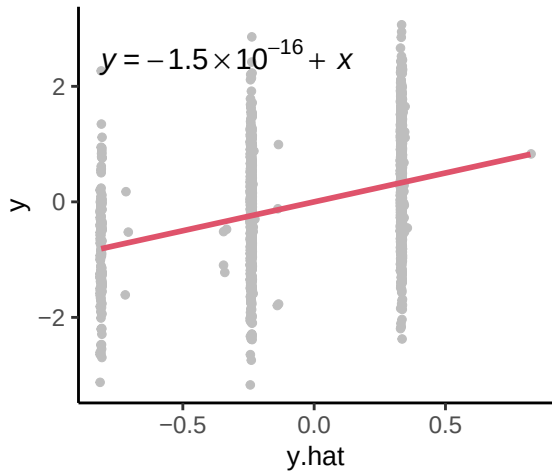
```
## Analysis of Variance Table
```

```
##
```

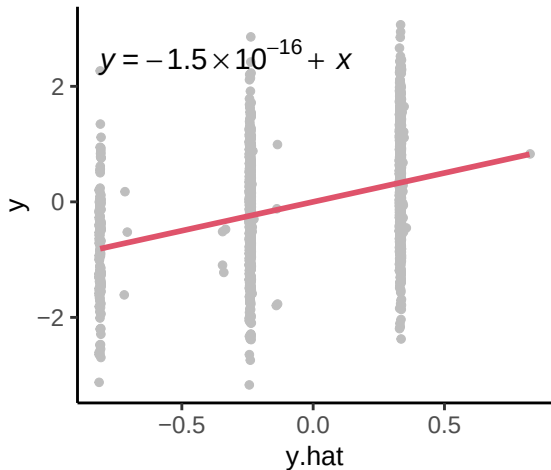
```
## Response: Y
```

```
##           Df  Sum Sq Mean Sq F value    Pr(>
```

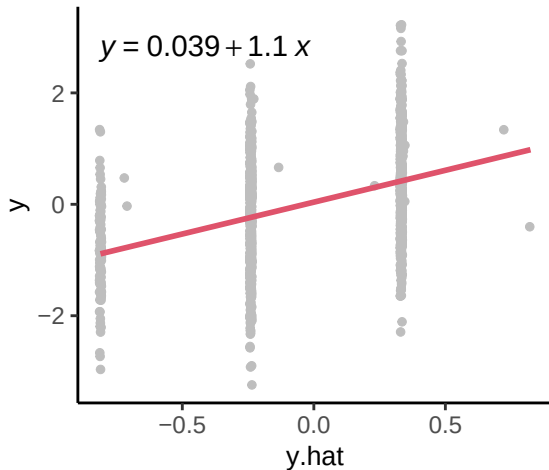

Train (N = 1500)



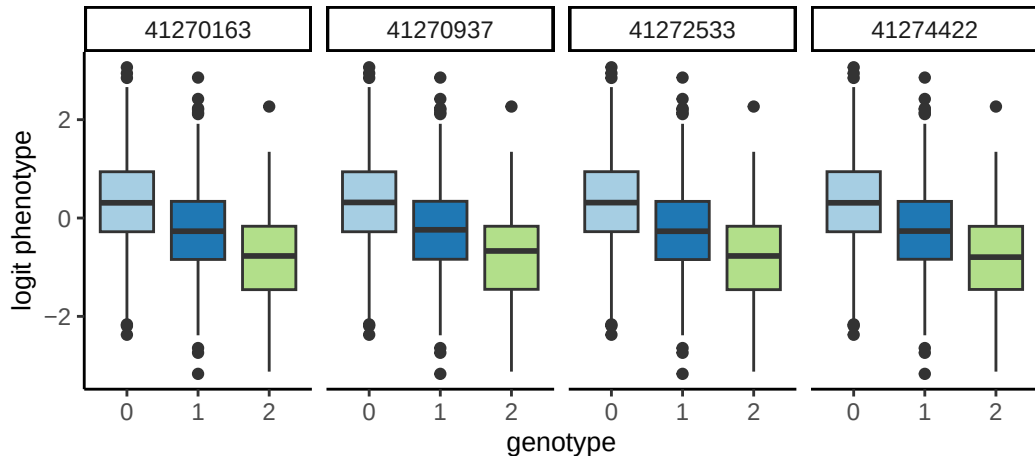
Train (N = 1500)



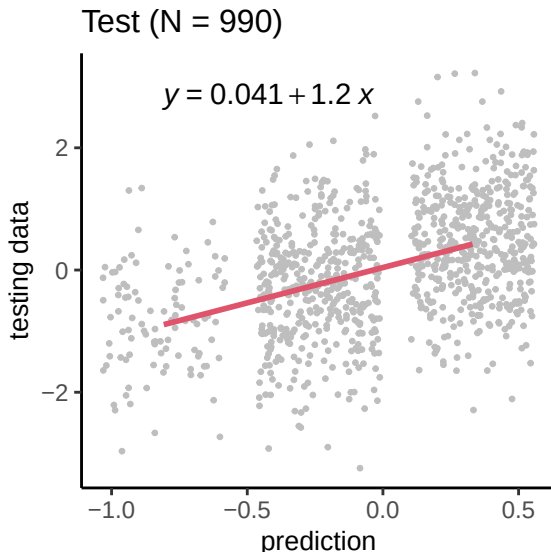
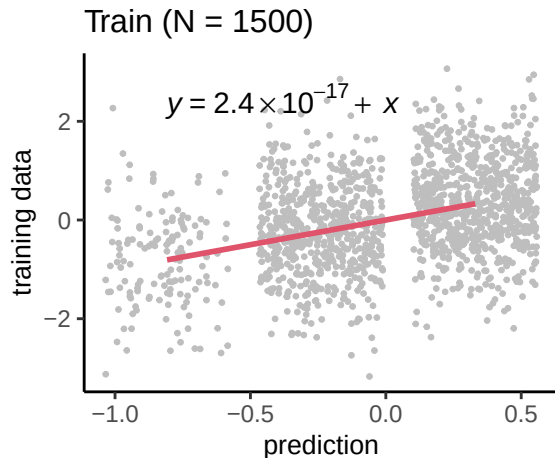
Test (N = 990)



Do we need all of these GWAS variants?



Just one of the GWAS variants works fine



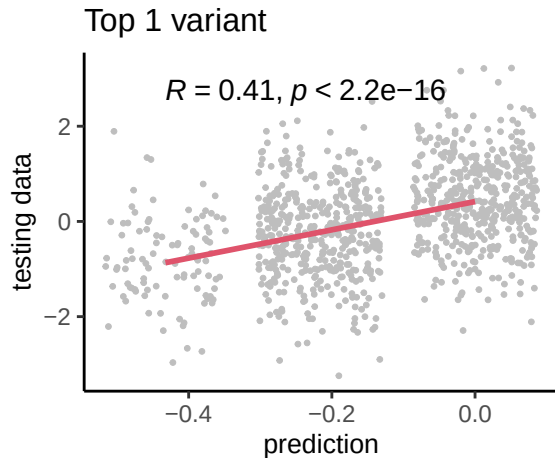
Simply add more variables?

Let's do some experiments:

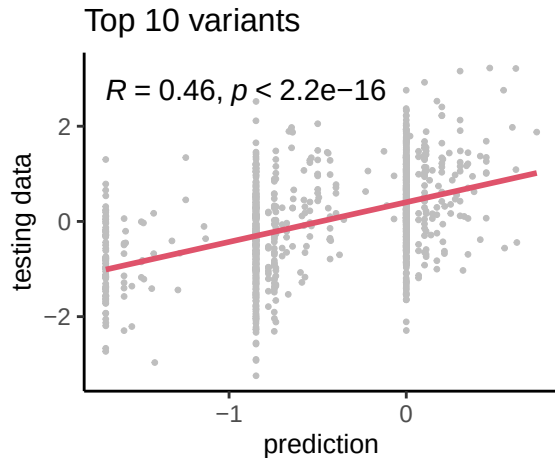
- 1 Pick top K variants (ordering by GWAS, $p_{(1)} < p_{(2)} < \dots$)
- 2 Take GWAS effect sizes β (the mean difference between case vs. control)
- 3 Predict by a linear combination of these effects:

$$\sum_{j=1}^K X_{i(j)} \beta_{(j)}$$

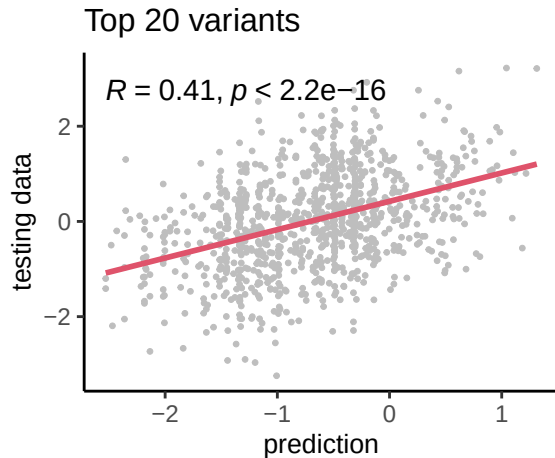
PGS is a variable selection problem



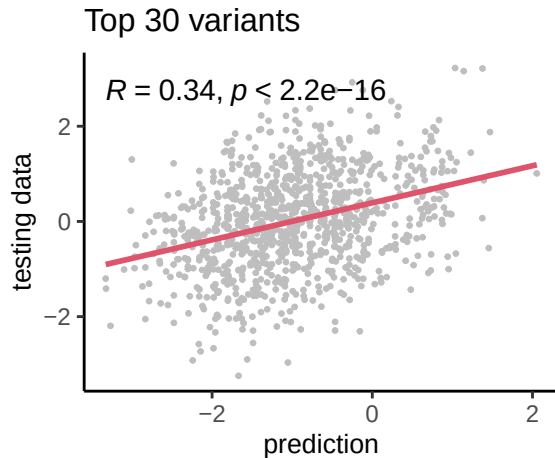
PGS is a variable selection problem



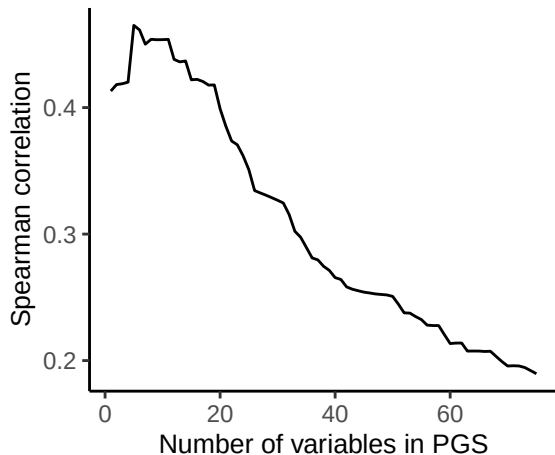
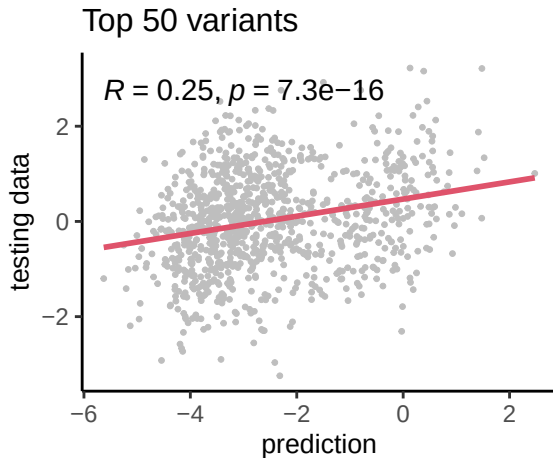
PGS is a variable selection problem



PGS is a variable selection problem



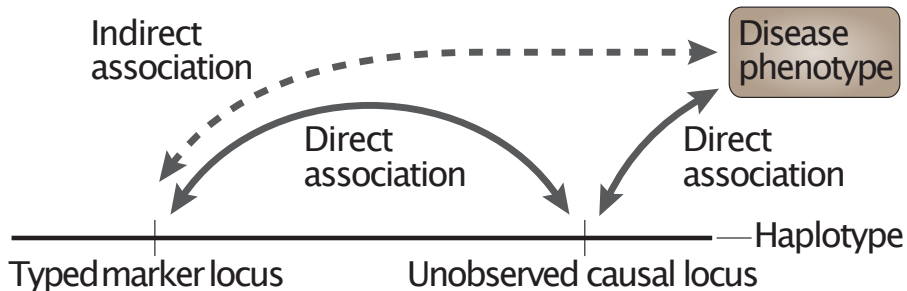
PGS is a variable selection problem



Today's lecture

- 1 What is GWAS (review and pick up the leftover)?
- 2 What is a polygenic score model?
- 3 What are the statistical challenges in PGS estimation?**
- 4 Statistical fine-mapping to handle LD structures

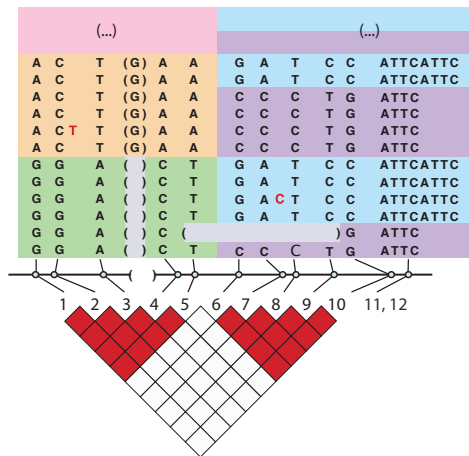
GWAS statistics only roughly tag causal variants



Balding et al. Nature Review Genetics (2006)

Covariance between variants obfuscates GWAS interpretation

- Linkage disequilibrium (LD) block
- The result of recombination events throughout generations



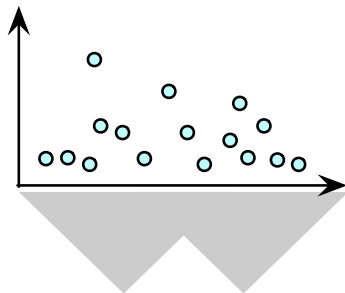
A typical setting of PGS training

PGS training

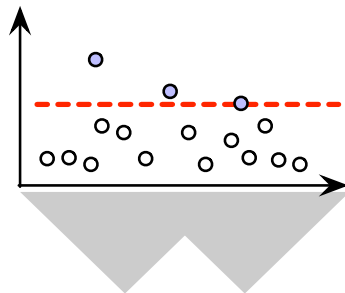
- Input:
 - 1 GWAS summary statistics (p-values, univariate effect sizes)
 - 2 $n \times p$ genotype matrix X , where $X_{ij} \in \{0, 1, 2\}$
 - 3 $n \times 1$ phenotype vector y
- Output:
 - Polygenic effect sizes $\hat{\beta}_1, \dots, \hat{\beta}_p$
- Objective:
 - We want to predict unseen Y_i^* by $f(X^*; \hat{\beta})$ as accurately as possible
 - We want $\beta_j = 0$ as many as possible; only some $\beta_j \neq 0$.

Common strategies to deal with LD structures

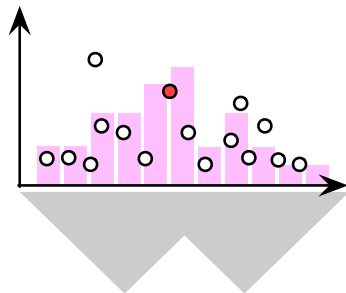
Strategy 1. Just use them all



Strategy 2. Pruning/clumping

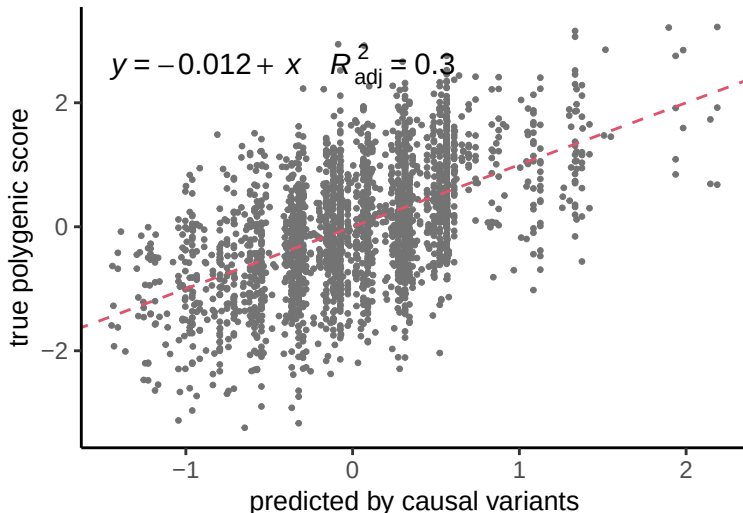


Strategy 3. Fine-mapping



True causal variables explain a large fraction of variation

```
.lm <- lm(.sim$y.q ~ .sim$x[, .sim$causal, drop = FALSE] - 1)
```



Today's lecture

- 1 What is GWAS (review and pick up the leftover)?
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- 4 Statistical fine-mapping to handle LD structures**

Classical statistics does not really help

- Classical variable selection by univariate (one-by-one) tests will not work for a $p \gg n$ regression problem
- Especially if we have col-linearity in the design matrix X
- also know as Linkage Disequilibrium (LD) in GWAS/PGS

Variable selection in high-dim. matrix ($n \ll p$)

Regression = projecting y on to columns $\{\mathbf{x}_j : j \in [p]\}$,

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \beta_1 \begin{pmatrix} X_{11} \\ X_{21} \\ \vdots \\ X_{n1} \end{pmatrix} + \dots \beta_p \begin{pmatrix} X_{1p} \\ X_{2p} \\ \vdots \\ X_{np} \end{pmatrix}.$$

Variable selection = column selection.

- Intuitive idea : choose the best combination of variables. $\rightarrow 2^p$ choices.

Variable selection in high-dim. matrix ($n \ll p$)

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- Alternative idea : make as many β_j 's nearly zero values.

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Variable selection = column selection.

- Intuitive idea : choose the best combination of variables. $\rightarrow 2^p$ choices.
- Alternative idea : make as many β_j 's nearly zero values.
- What prior does: penalize $|\beta_j| > 0$ so that only the strong enough variables take non-zero values.

Lasso, a linear regression with Laplace prior (L1)

Prior distribution

$$p(\theta) = \text{Laplace}(\theta|\lambda) \propto \exp(-\lambda\|\theta\|_1)$$

where

$$\|\theta\|_1 = \sum_{j=1}^p |\theta_j|, \text{ L1-norm.}$$

Maximize

$$\ln p(y|X, \theta) + \ln p(\theta|\lambda) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i\theta)^2 - \lambda\|\theta\|_1$$

Minimize L_1 -regularized error

glmnet solves this regularized optimization problem

Goal (by variable-by-variable updates):

$$\min_{\beta} \quad \overbrace{(\mathbf{y} - X\beta)^\top (\mathbf{y} - X\beta)}^{\text{RSS}} + \underbrace{\lambda\alpha\|\beta\|_1}_{\text{variable selection}} + \underbrace{\lambda(1-\alpha)\|\beta\|_2}_{\text{shrinkage}}$$

glmnet solves this regularized optimization problem

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$$\min_{\beta} \quad \overbrace{(\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)}^{\text{RSS}} + \underbrace{\lambda\alpha\|\beta\|_1}_{\text{variable selection}} + \underbrace{\lambda(1-\alpha)\|\beta\|_2}_{\text{shrinkage}}$$

For each β_j ,

$$\hat{\beta}_j^{\text{glmnet}} \leftarrow \frac{S\left(\sum_{i=1}^n X_{ij}(y_i - \hat{y}_i^{(-j)}), \lambda\alpha\right)}{\sum_{i=1}^n X_{ij}^2 + \lambda(1-\alpha)}$$

Friedman et al., Regularization Paths for Generalized Linear Models via Coordinate Descent (2010)

glmnet solves this regularized optimization problem

Goal (by variable-by-variable updates):

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For each β_j ,

$$\hat{\beta}_j \leftarrow \frac{\text{threshold} \left(\begin{array}{c} \text{residual w/o the variable } \beta_j \\ \sum_{i=1}^n X_{ij} \quad \overbrace{(y_i - y_i^{(-j)})} \end{array}, \lambda\alpha \right)}{\sum_{i=1}^n X_{ij}^2 + \underbrace{\lambda(1-\alpha)}_{\text{shrinkage}}}$$

Running glmnet for polygenic risk prediction of OGD

```
glm.cv.out <- glmnet::cv.glmnet(X, Y, nfolds=5, alpha=1)
lambda.cv <- glm.cv.out$lambda.min
glm.out <- glmnet::glmnet(x=X, y=Y, lambda = lambda.cv, alpha=1)
```

```
head(glm.out$beta)
```

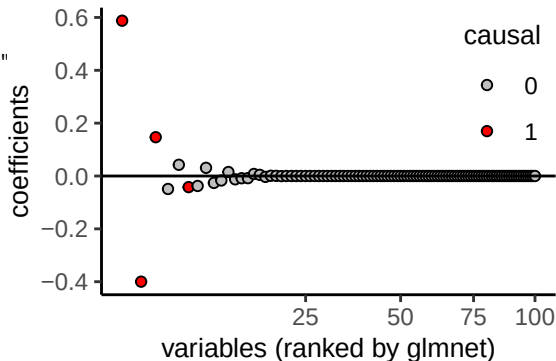
```
## 6 x 1 sparse Matrix of class "dgCMatrix"
##      s0
## 566875 .
## 752721 .
## 754105 .
## 754182 .
## 754334 .
## 756604 .
```

Running glmnet for polygenic risk prediction of OGD

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```
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```

```
## 6 x 1 sparse Matrix of class "dgCMatrix"
##          s0
## 566875 .
## 752721 .
## 754105 .
## 754182 .
## 754334 .
## 756604 .
```



Sum of Single Effect (SuSiE) regression

```
library(susieR)
susie.out <- susie(X, Y)
```

$$y = \sum_{l=1}^L \underbrace{\sum_j x_j \alpha_j^{(l)} \beta_j^{(l)}}_{\text{layer-by-layer}} + \epsilon$$

probabilistic selection single variant effect

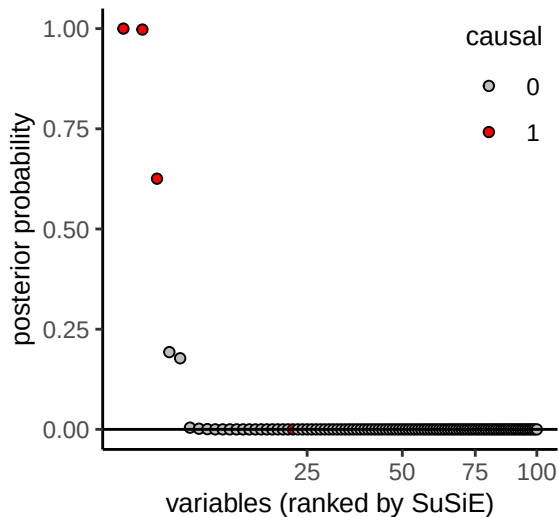
where $\sum_{j=1}^p \alpha_j^{(l)} = 1$ for each layer l .

Wang .. Stephens, *Journal of the Royal Statistical Society* (2020)

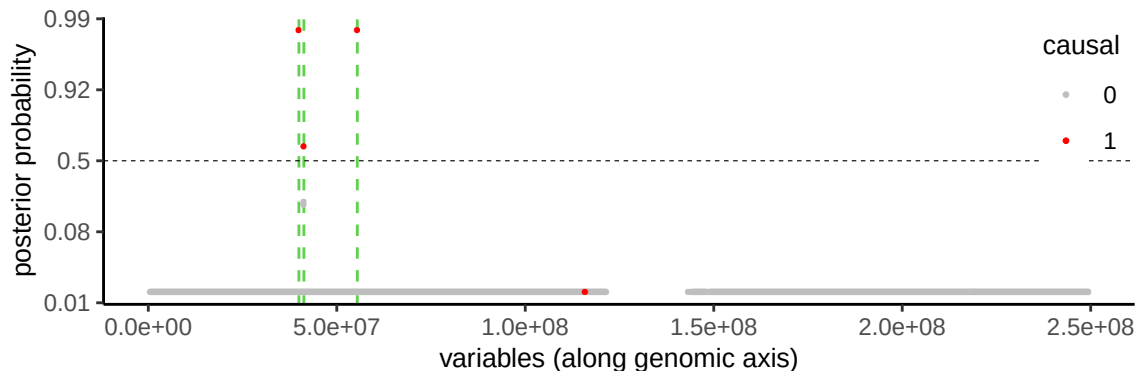
Ideas behind SuSiE

- 1 Estimating univariate effects is easy (GWAS)
- 2 Many variants can show similar effects (LD)
- 3 Let's weight variants probabilistically
- 4 Regress out the probabilistically reweighted effects
- 5 Repeat 1-4.

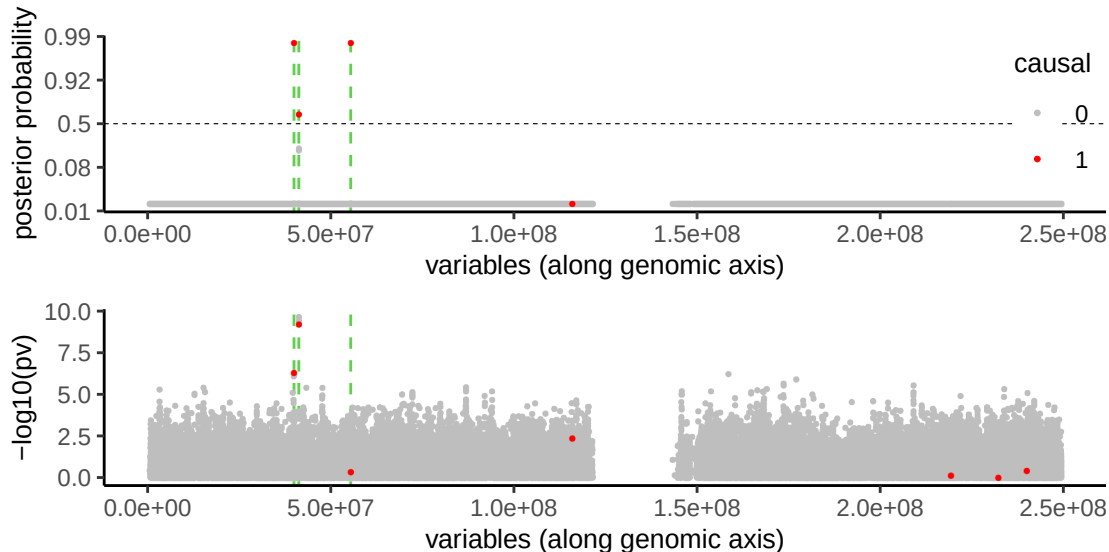
SuSiE identifies top causal variants



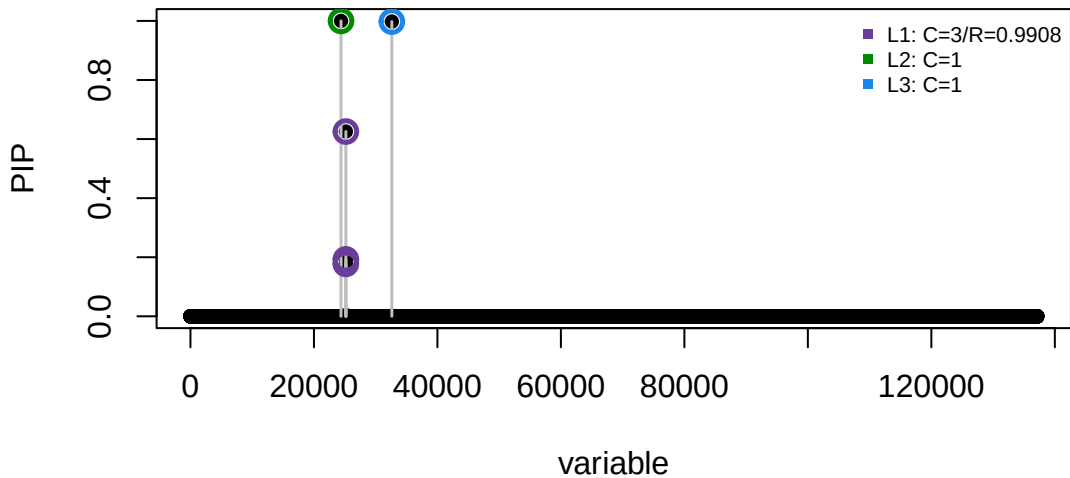
SuSiE can avoid the col-linearity (LD) problem



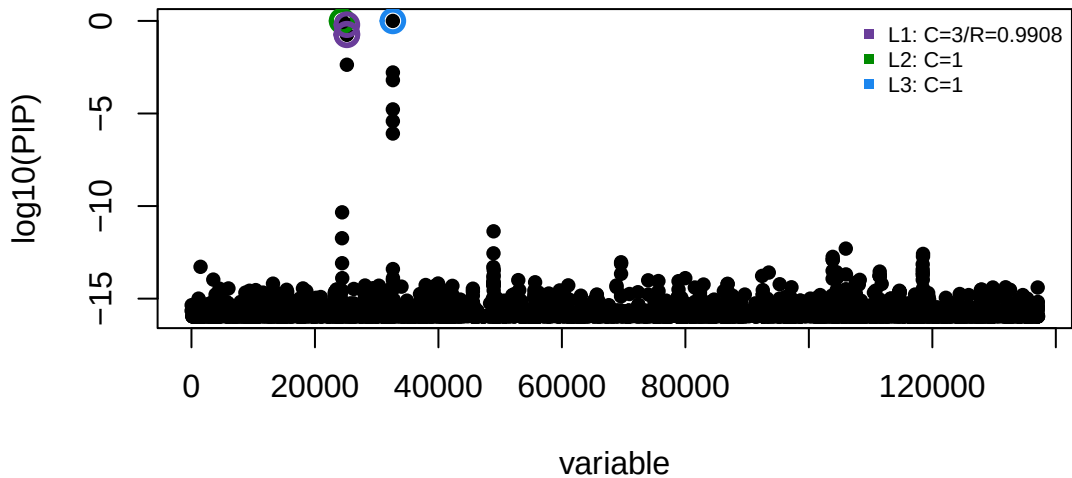
SuSiE can avoid the col-linearity (LD) problem



SuSiE can avoid the col-linearity (LD) problem

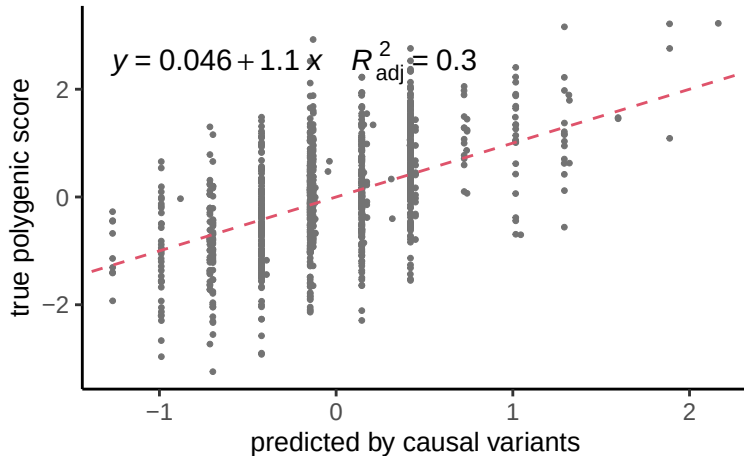


SuSiE can avoid the col-linearity (LD) problem

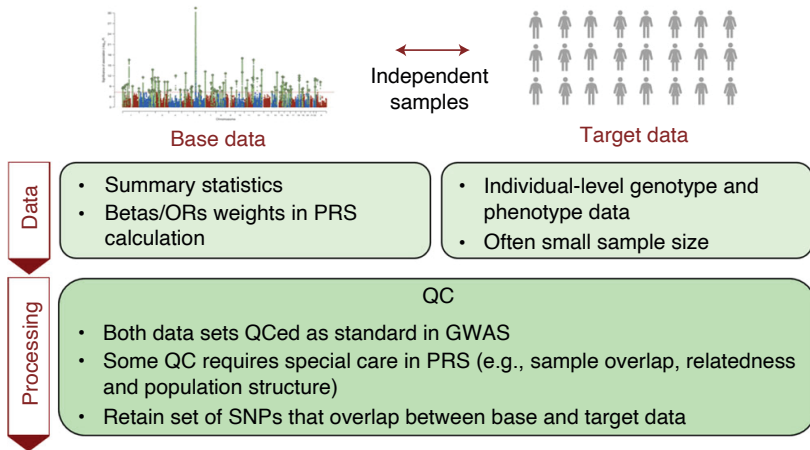


SuSiE+PGS accurately predict the held-out data

```
y.hat <- predict.susie(susie.out, x.test)
```

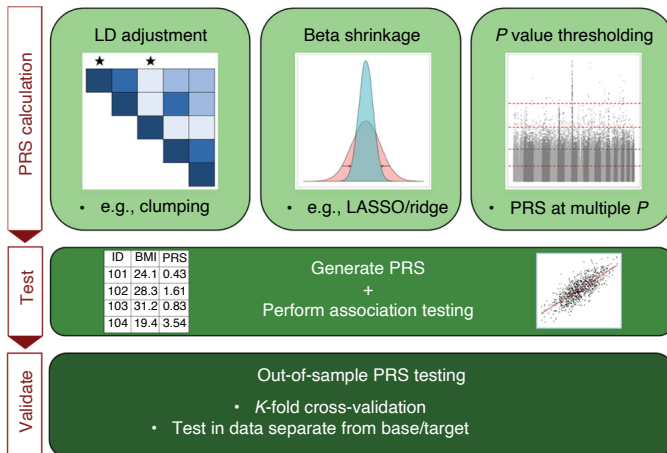


A general workflow of PGS estimation



Choi .. O'Reilly, Nature Protocol, (2020)

A general workflow of PGS estimation



Choi .. O'Reilly, Nature Protocol, (2020)

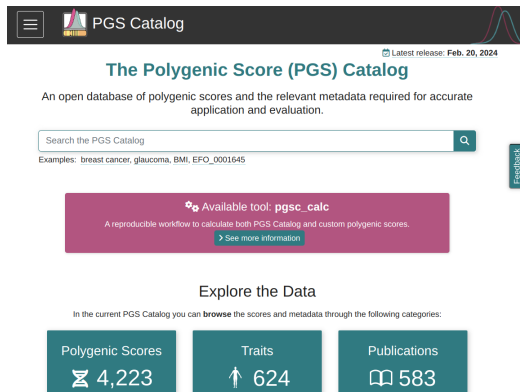
A proposed workflow of PGS estimation

- 1 Take GWAS summary statistics and target population genotype X
- 2 Run SuSiE² chromosome by chromosome to select variables
- 3 Predict PGS per chromosome and aggregate them

²SuSiE can run with summary statistics only

PGS Catalog: No need to train from scratch

<https://www.pgscatalog.org/>



The screenshot shows the PGS Catalog website. At the top is a dark header with a menu icon, the PGS Catalog logo, and a bell icon. Below the header, it says "The Polygenic Score (PGS) Catalog" and "An open database of polygenic scores and the relevant metadata required for accurate application and evaluation." There is a search bar with the placeholder text "Search the PGS Catalog" and a magnifying glass icon. Below the search bar, it says "Examples: breast cancer, glaucoma, BMI, EFO_0001645". A purple box highlights the available tool "pgsc_calc" with a description and a link to see more information. Below this is the "Explore the Data" section, which states "In the current PGS Catalog you can browse the scores and metadata through the following categories:". There are three teal boxes: "Polygenic Scores" with a bar chart icon and the number 4,223, "Traits" with a person icon and the number 624, and "Publications" with a book icon and the number 583. A vertical "Feedback" button is on the right side.

PGS Catalog

Latest release: Feb. 20, 2024

The Polygenic Score (PGS) Catalog

An open database of polygenic scores and the relevant metadata required for accurate application and evaluation.

Search the PGS Catalog

Examples: breast cancer, glaucoma, BMI, EFO_0001645

Available tool: **pgsc_calc**

A reproducible workflow to calculate both PGS Catalog and custom polygenic scores.

[See more information](#)

Explore the Data

In the current PGS Catalog you can **browse** the scores and metadata through the following categories:

Polygenic Scores	Traits	Publications
4,223	624	583

Advanced topics that we might consider next lecture

- Summary-based GWAS, post-GWAS analysis
- Mendelian Randomization and other causal inference techniques